higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

MARKING GUIDELINE

NATIONAL CERTIFICATE (VOCATIONAL)

INSTRUMENTATION TECHNOLOGY
NQF LEVEL 4

18 NOVEMBER 2019

This marking guideline consists of 7 pages.

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SECTION A

QUESTION 1

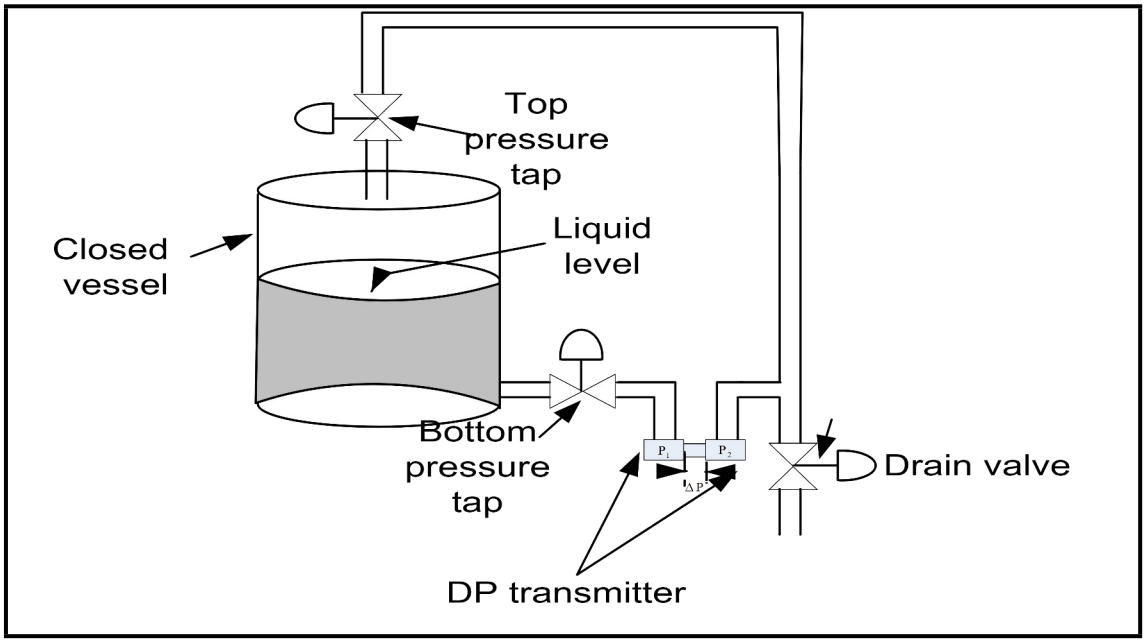
1.1	1.1.1	D		
	1.1.2	D		
	1.1.3	D		
	1.1.4	C		
	1.1.5	A		
			(5 × 2)	(10)
1.2	1.2.1	G		
	1.2.2	F		
	1.2.3	A		
	1.2.4	C		
	1.2.5	H		
			(5 × 1)	(5)
1.3	1.3.1	True		
	1.3.2	True		
	1.3.3	False		
	1.3.4	False		
	1.3.5	True		
			(5 × 1)	(5)
1.4	1.4.1	Mass		
	1.4.2	Pneumatic signal		
	1.4.3	Hydrostatic		
	1.4.4	Weight		
	1.4.5	Turbidity		
			(5 × 2)	(10)
				[30]

TOTAL SECTION A: 30

SECTION B

QUESTION 2

2.1



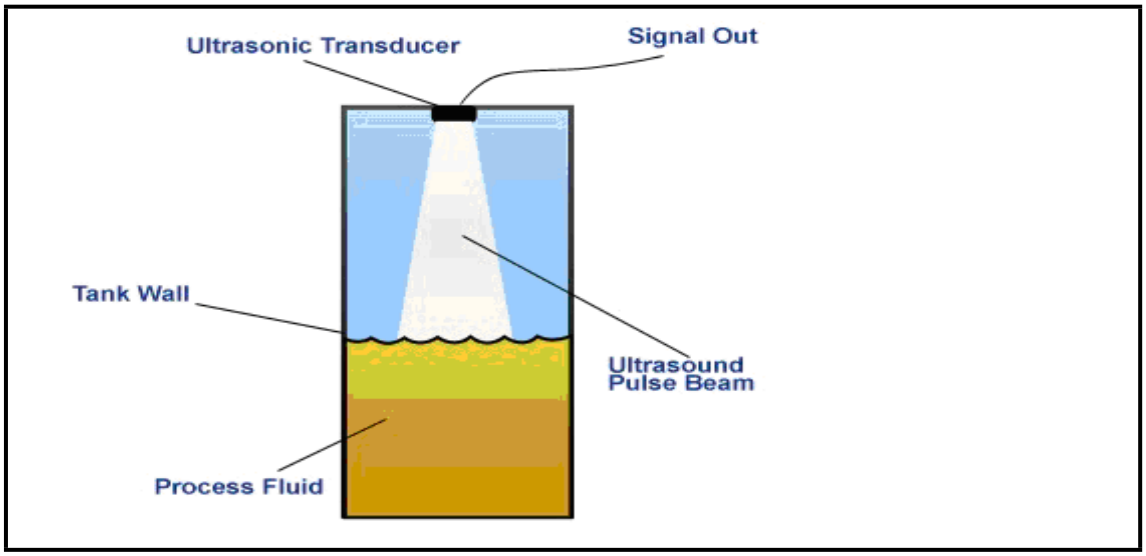
(6)

2.2

- If a high viscosity or liquid with solids in suspension is to be measured, a standard DP transmitter cannot be used as the process port connected to the bottom of the vessel could become blocked.✓✓
- A flanged DP or a transmitter with a remote seal should be used in these circumstances.✓

(3)

2.3



(5)

2.4 The level sensor located at the top of the tank sends out the sound waves in the form of bursts✓ in the downward direction to the fluid in the tank under level measurement.✓ As soon as the directed sound waves hit the surface of the fluid, sound echoes are reflected✓ and returned to the sensor.✓
The time taken by the sound wave to return✓ is directly proportional to the distance between the piezoelectric sensor and the material in the tank.✓ This time duration is measured by the sensor which is then further used to calculate the level of liquid in the tank.✓ (7)

2.5
- Normal installation
  The transmitter is mounted level with minimum level tap.
- Elevated zero installation
  The transmitter is mounted above the high-pressure tap.
- Suppressed zero installation
  The transmitter is mounted below the high-pressure tap. (3 × 2) (6)

2.6 The system is based on the reflection of an electromagnetic pulse that is directed on a sensor rod/cable and reflected by filling material.✓
The electronics integrated into the sensor determines the filling state from the echo time of the pulse and shows this in the display.✓

The electromagnetic pulse is sent out, reflected on the filling material and received again.✓ The echo time of the pulse is proportional to the distance from the surface of the medium. The measurement procedure and the accuracy of the measurement are independent of pressure,✓ temperature,✓ vapour,✓ dust,✓ foam,✓ viscosity,✓ conductivity and pH value.✓
(Any 6 × 1) (6)
[33]

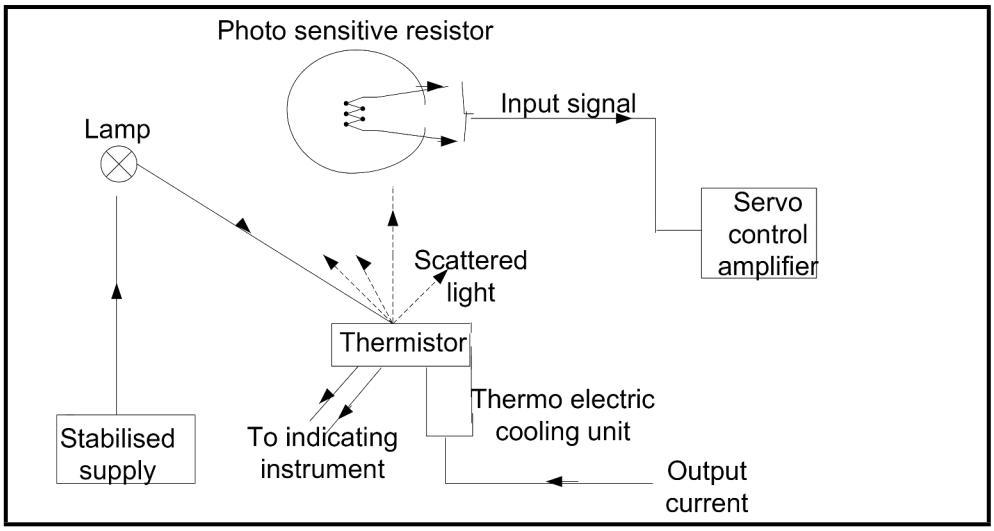
QUESTION 3

3.1
- Rotating concentric cylinder viscometer
- Falling piston viscometer
- Vibrating-reed viscometer
- Variable area viscometer
- Hand-held viscometer (5)

3.2 3.2.1
- Gases which are magnetised in the direction of the applied magnetic field,✓ i.e. attracted to the strongest part of the magnetic field✓ are called paramagnetic gases.
- Gases which are repelled by a magnetic field✓ or magnetised in the opposite direction✓ or which move to the weaker part of the magnetising field are called diamagnetic gases.✓ (5)

3.2.2
- Heating plants
- Steam power plants (2)

3.3 3.3.1



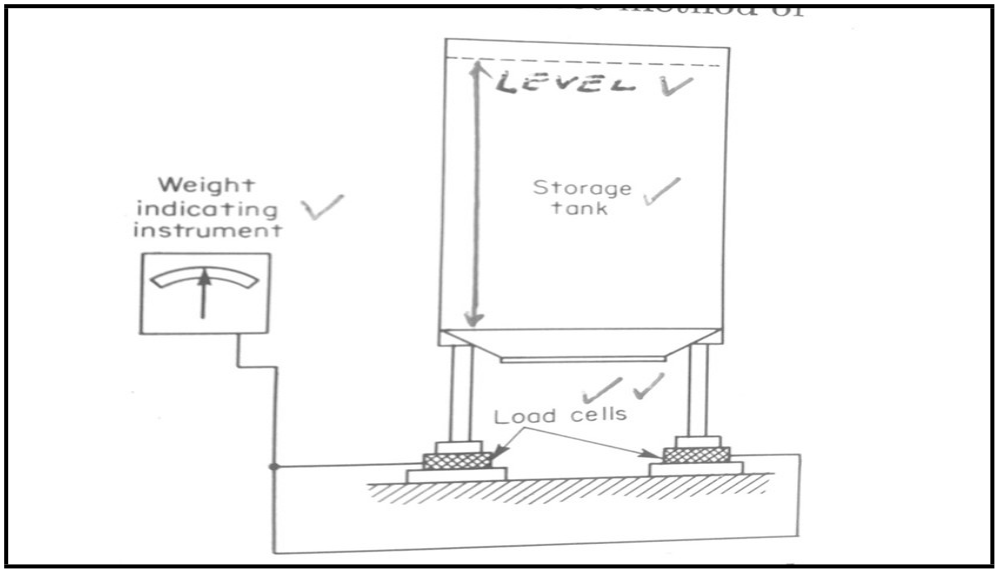
(8)

3.3.2 A light beam from a lamp will be directed in the direction of the surface.✓ A photo sensitive resistor is placed at a point where the radiated light beam will fall✓ after passing through the dew drops formed on the mirror,✓ due to the cooling of the mirror.✓ If no dew drops are present, ✓ the light beam will not be picked up by the photo sensitive resistor.✓ The more dew drops, the more reflected light will be picked up.✓ This signal will be sent to the cooler to switch off so that the temperature will fluctuate around the dew point.✓ A thermistor is used to determine the temperature of the mirror.✓ This temperature will be the dew point temperature of the substance.✓

(10)
[30]

QUESTION 4

4.1 4.1.1



(5)

4.1.2 The load cells are specially designed and constructed units containing strain gauges.✓ The strain gauges provide a measurable electrical output✓ proportional to the stress applied by the weight acting upon the load cells.✓ As the pressure on the load cells due to variations in the weight of the material inside the tank changes,✓ the electrical resistances of the strain gauges also change.✓ The strain gauge is connected to a measuring bridge✓ containing an electrical meter calibrated in units of level measurement.✓ (7)

4.2
- Carry out visual inspections of the cable for damage in the event that problems may arise.
- Keep the plug clean and dry at all times.
- Avoid carrying the load cell by the cable.
- If the cable is damaged, the cable should be repaired by cutting and splicing.
- All splices should be soldered, mechanically strong, well-insulated and protected from dirt and moisture.
- Check the load-cell wiring. (Any 5 × 1) (5)

4.3 The types of load cells are classified based on three criteria:

(1) Working principle of a load cell✓✓
- Compression principle
- Tensile-based working
- Universal
- Hollow
- Shear based

(2) Construction of a load cell ✓✓
- S-type construction load cell
- Load button types
- Single-ended shear beam
- Double-ended shear beam load cell
- Single-column and multicolumn load cell
- Pancake load cell
- Diaphragm/membrane
- Torsion ring load cell

(3) Electrical properties of a load cell✓✓
- Analogue and digital property-based load cell
- Resistance- and capacitance-based
- Piezoelectric- and wireless-based load cell (3 × 2) (6)

4.4
- Calibration is affected by changes in the rigidity of the flow tubes
- Changes in the temperature will affect tube rigidity
- Pressure changes
- Degradation of the flow tube
- Cracking, coating , erosion or corrosion of the flow tube (Any 4 × 1) (4)

[27]

QUESTION 5

5.1 5.1.1 Fail safe open means that an input signal is needed in order for the safety device to stay close.✓ Should the input signal be removed the safety device will open✓

5.1.2 Fail safe closed means that an input-signal must be present in order for the safety device to stay open.✓ Should the input signal be removed, the safety device will close✓

(2 × 2) (4)

5.2
- Monitors and controls a process
- Meets product specifications
- Reduces waste
- Reduces the time required to produce a product
- Ensures that process operates at its full potential (5)

5.3
- A human-machine interface or HMI
- A computer system
- Remote terminal units (RTUs)
- Programmable logic controller (PLCs)
- Communication infrastructure (5)

5.4 5.4.1 Temperature element
5.4.2 Temperature recording controller

(2 × 3) (6)

5.5 5.5.1 Cascade control loop (2)

5.5.2 Controller 1 (primary controller) compares the temperature of the outflow✓ with the temperature set point.✓ Should there be an error✓ it becomes the input to the controller 2 (secondary controller), which will be compared with the steam flow rate.✓ The output of controller 2 regulates the valve opening✓ in order to increase or decrease the steam supply,✓ resulting in an increase or decrease in the temperature of water in the tank✓ to meet the temperature set point.✓ (8)

[30]

TOTAL SECTION B: 120
GRAND TOTAL: 150

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